

Bansilal Ramnath Agarwal Charitable Trust's
Vishwakarma Institute of Technology, Pune-37

(An autonomous Institute of Savitribai Phule Pune University)



Department of Multidisciplinary
Engineering

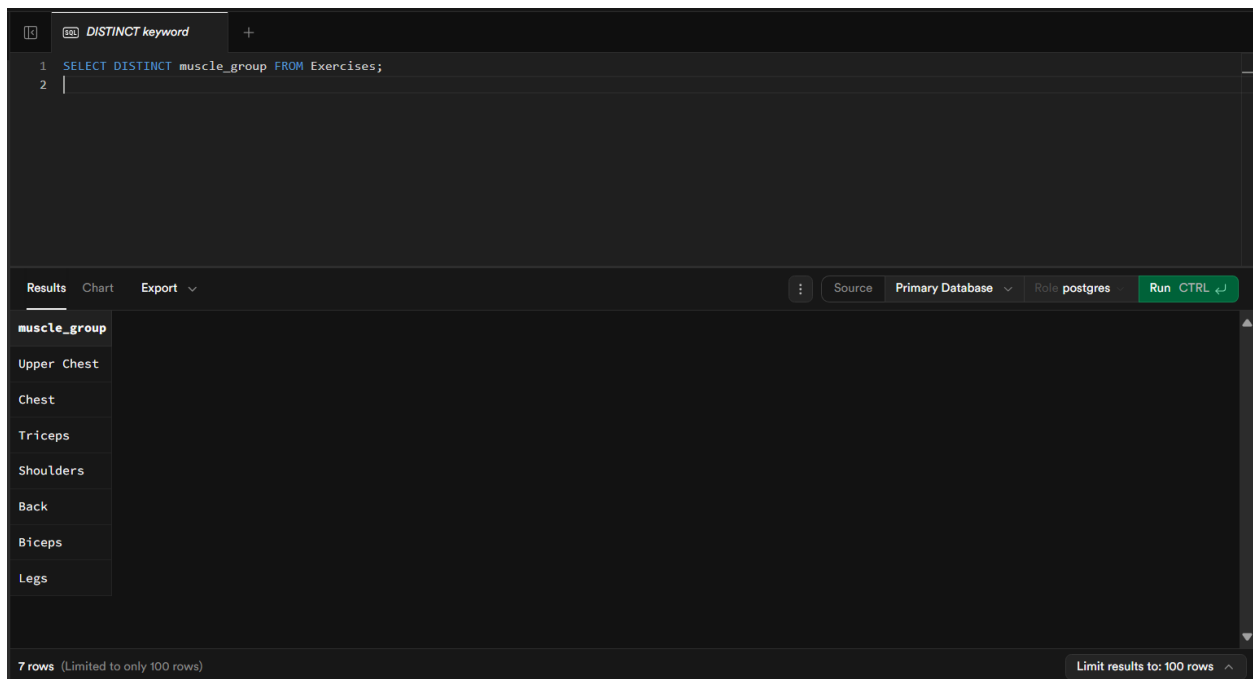
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Title:

Distinct keyword, where clause, on clause, having clause, group by clause with aggregate functions, set operations, nested query and correlated query

Description: In this assignment, we explored key SQL concepts used for querying databases. The DISTINCT keyword helps remove duplicate values, while the WHERE clause filters rows based on conditions. The ON clause specifies join conditions between tables. The GROUP BY clause groups records for analysis, and with aggregate functions like COUNT or AVG, it summarizes data, while the HAVING clause filters these groups. Set operations such as UNION, INTERSECT, and EXCEPT combine or compare results of multiple queries. Nested queries (subqueries) are used inside another query for filtering or calculation, while correlated queries depend on values from the outer query for execution. Together, these concepts allow efficient data retrieval and analysis.

Screenshots/Output:



The screenshot shows a SQL IDE interface. At the top, there's a tab labeled "DISTINCT keyword". Below it, the SQL query is entered in a text area:

```
1 SELECT DISTINCT muscle_group FROM Exercises;  
2 |
```

Below the query editor, there's a toolbar with "Results", "Chart", and "Export" buttons. To the right of these buttons are "Source", "Primary Database", "Role postgres", and a green "Run CTRL ↵" button. Below the toolbar, the results are displayed in a table with the column header "muscle_group". The table contains the following rows:

muscle_group
Upper Chest
Chest
Triceps
Shoulders
Back
Biceps
Legs

At the bottom left, it says "7 rows (Limited to only 100 rows)". At the bottom right, it says "Limit results to: 100 rows ^".

WHERE clause

1

2

SELECT name, muscle_group FROM Exercises WHERE muscle_group = 'Chest';

Results

Chart

Export

Source

Primary Database

Role postgres

Run CTRL ↵

name	muscle_group
Chest Fly	Chest
Push Up	Chest
Push Up	Chest

3 rows (Limited to only 100 rows)

Limit results to: 100 rows

ON clause

1

2

3

4

5

SELECT e.name, eq.equipment_name
FROM Exercises e
JOIN Available_Equipments eq
ON e.equipment_id = eq.equipment_id;

Results

Chart

Export

Source

Primary Database

Role postgres

Run CTRL ↵

name	equipment_name
Bicep Curl	Adjustable Dumbbells
Tricep Pushdown	Cable Machine
Shoulder Press	Adjustable Dumbbells
Lat Pulldown	Cable Machine
Leg Press	Squat Rack
Chest Fly	Adjustable Dumbbells
Bench Press	Barbell
Push Up	Barbell

10 rows (Limited to only 100 rows)

Limit results to: 100 rows

GROUP BY clause with aggregate functions

```
1 SELECT muscle_group, COUNT(*) AS total_exercises
2 FROM Exercises
3 GROUP BY muscle_group;
4
```

ResultsChartExport

SourcePrimary DatabaseRole postgresRun CTRL ↵

muscle_group	total_exercises
Upper Chest	1
Chest	3
Triceps	1
Shoulders	2
Back	1
Biceps	1
Legs	1

7 rows (Limited to only 100 rows)Limit results to: 100 rows ^

HAVING clause

```
1 SELECT muscle_group, COUNT(*) AS total_exercises
2 FROM Exercises
3 GROUP BY muscle_group
4 HAVING COUNT(*) > 2;
5
```

ResultsChartExport

SourcePrimary DatabaseRole postgresRun CTRL ↵

muscle_group	total_exercises
Chest	3

1 row (Limited to only 100 rows)Limit results to: 100 rows ^

Set Operations (UNION)

1

SELECT name FROM Exercises WHERE equipment_id IS NOT NULL

2

UNION

3

SELECT name FROM Exercises WHERE muscle_group = 'Legs';

4

Results

Chart

Export

Source

Primary Database

Role postgres

Run CTRL ↵

name
Squat
Leg Press
Push Up
Bicep Curl
Shoulder Press
Bench Press
Tricep Pushdown
Chest Fly

9 rows (Limited to only 100 rows)Limit results to: 100 rows

Set Operations (INTERSECT)

1

SELECT name FROM Exercises

2

INTERSECT

3

SELECT equipment_name FROM Available_Equipments;

Results

Chart

Export

Source

Primary Database

Role postgres

Run CTRL ↵

Success. No rows returned

0 row (Limited to only 100 rows)Limit results to: 100 rows

Set Operations (EXCEPT)

1

SELECT name FROM Exercises

2

EXCEPT

3

SELECT equipment_name FROM Available_Equipments;

4

|

Results

Chart

Export

Source

Primary Database

Role postgres

Run CTRL ↵

name

Squat

Leg Press

Push Up

Bicep Curl

Shoulder Press

Bench Press

Tricep Pushdown

Chest Fly

9 rows (Limited to only 100 rows)

Limit results to: 100 rows

Nested Query (Subquery)

1

SELECT name

2

FROM Exercises

3

WHERE equipment_id = (SELECT equipment_id FROM Available_Equipments WHERE equipment_name = 'Dumbbell');

4

Results

Chart

Export

Source

Primary Database

Role postgres

Run CTRL ↵

Success. No rows returned

0 row (Limited to only 100 rows)

Limit results to: 100 rows

Correlated Query

```
1 SELECT name, equipment_id
2 FROM Exercises e
3 WHERE equipment_id IN (
4     SELECT equipment_id
5     FROM Available_Equipments eq
6     WHERE eq.equipment_id = e.equipment_id
7 );
8
```

ResultsChartExport

SourcePrimary DatabaseRole postgresRun CTRL ↵

name	equipment_id
Bicep Curl	4
Tricep Pushdown	3
Shoulder Press	4
Lat Pulldown	3
Leg Press	2
Chest Fly	4
Bench Press	1
Push Up	1

10 rows (Limited to only 100 rows)Limit results to: 100 rows